UNIVERSITÀ DEGLI STUDI DI TRIESTE

Defensive distillation against Adversarial attacks for Deep neural networks

Advanced Topics in Machine Learning

Poornima Devi Krishnasamy Karthikeyan

https://arxiv.org/abs/1511.04508

Content

- Objective
- Dataset Description
- Implementation
  - Fast Gradient Sign attack (FGSM).
  - Baseline model
  - Baseline model in the presence of FGSN
- Distillation models
  - Distillation as defence
  - Implementation of distillation model with FGSM
- Results

Objective :

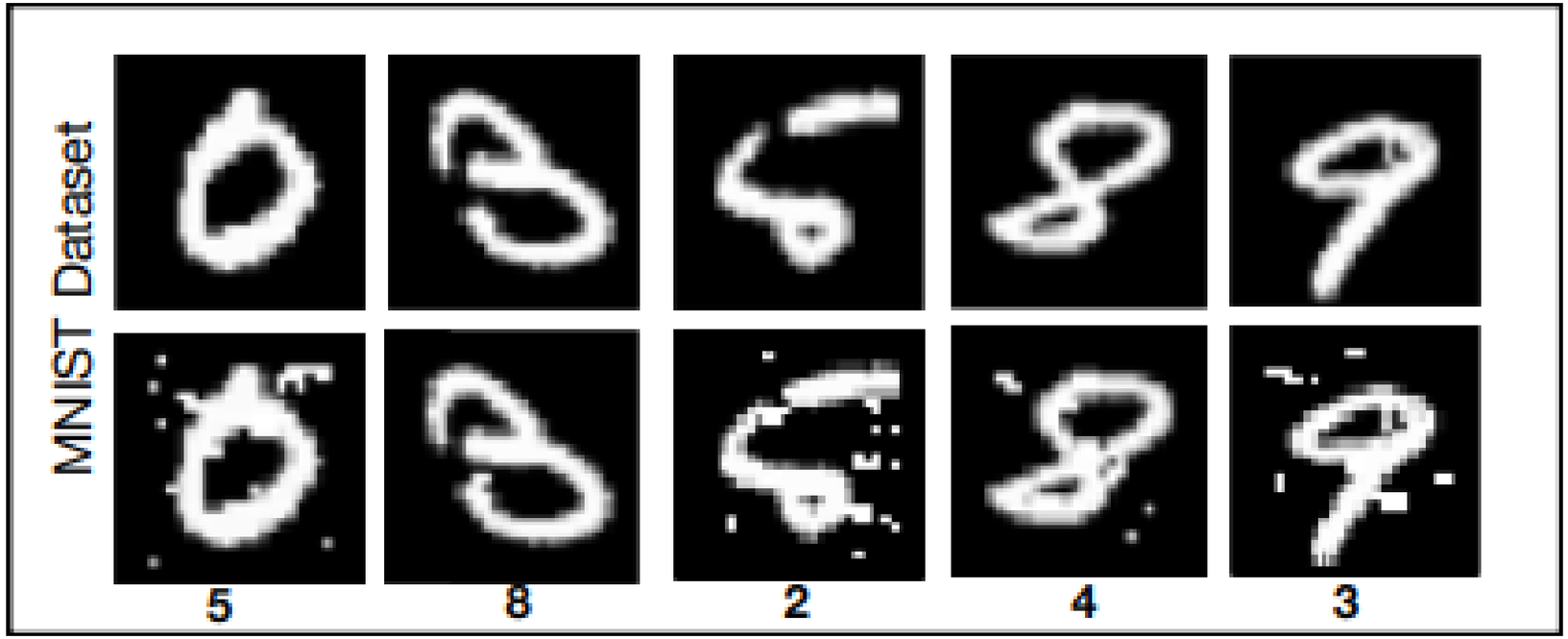
- **Problem** : Neural networks are vulnerable to small, imperceptible perturbations that can cause misclassification.
- **Goal** : Generate an adversarial attack and use a defensive strategy to make classification model robust.
  - Attack → Generate Fast Gradient Sign Method (FGSM).
    - Examined the attack impacts on the classification accuracy of the model
  - Defense → Defensive distillation
    - Train the student network F1 using soft labels from Teacher F → reduce gradients, encouraging the output to be nearly flat close to training examples → improves classification accuracy against some adversarial attacks during testing.

Dataset Description :

MNIST : Contains 70,000 images of handwritten digits (0–9).

- Train 50,000 images
- Valid 10,000 images
- Test 10,000 images

- Each image is 28×28 pixels, grayscale, and centered

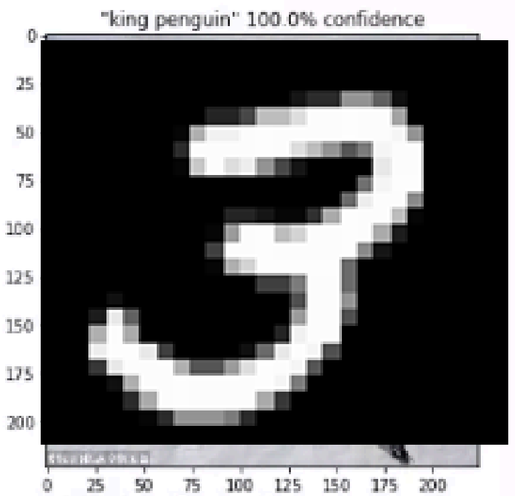


Implementation : Fast Gradient Sign Method

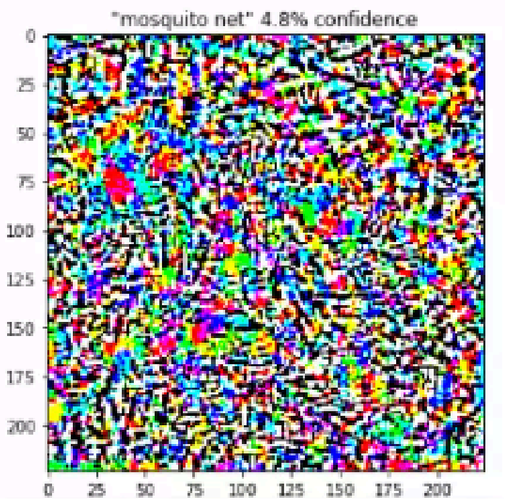
Attack setting :

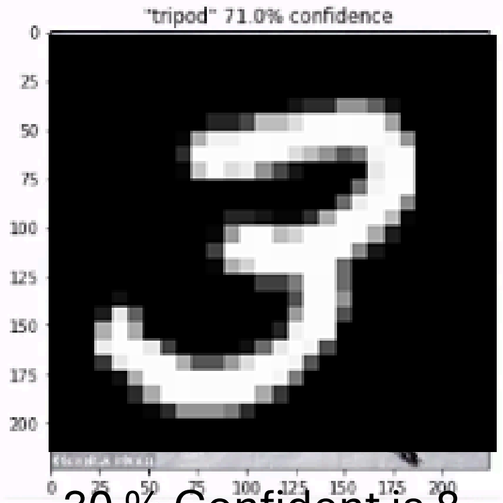
- White box attack ⇒ it requires knowledge of the model's architecture and parameters. Untargeted.

$$x \quad + \quad \varepsilon \ \mathrm{sign}(\nabla_x L(\theta, x, y)) \quad = \quad x_{\mathrm{adv}}$$



97 % Confident is 3





30 % Confident is 8

crafted to maximize the loss causes misclassification

ϵ controls the intensity of the noise

Larger ε → stronger attack → higher misclassification

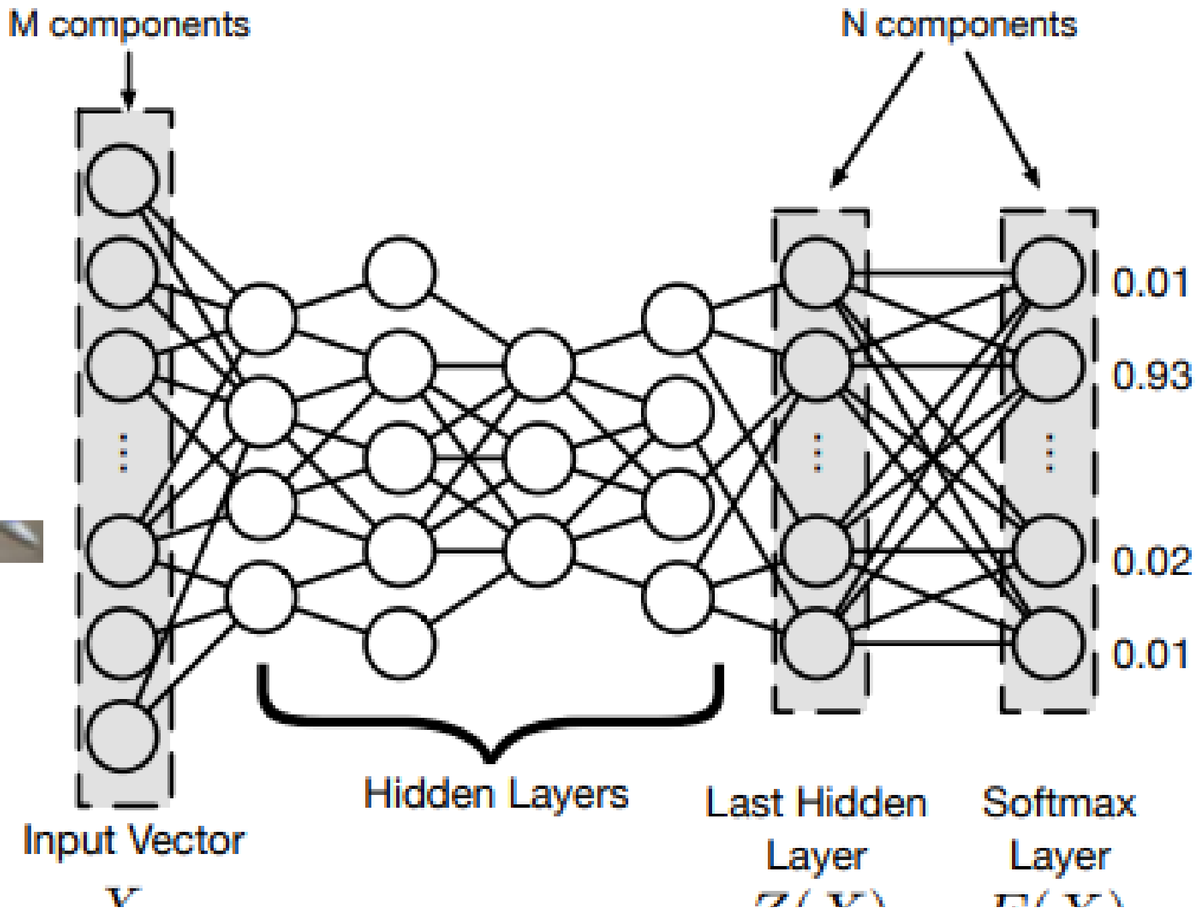
Goodfellow et al. in 2014 [3].

Implementation : Baseline model

At Training

Train set Images

Input



Test accuracy 97% with true test images

3

Objective: **minimize the loss**

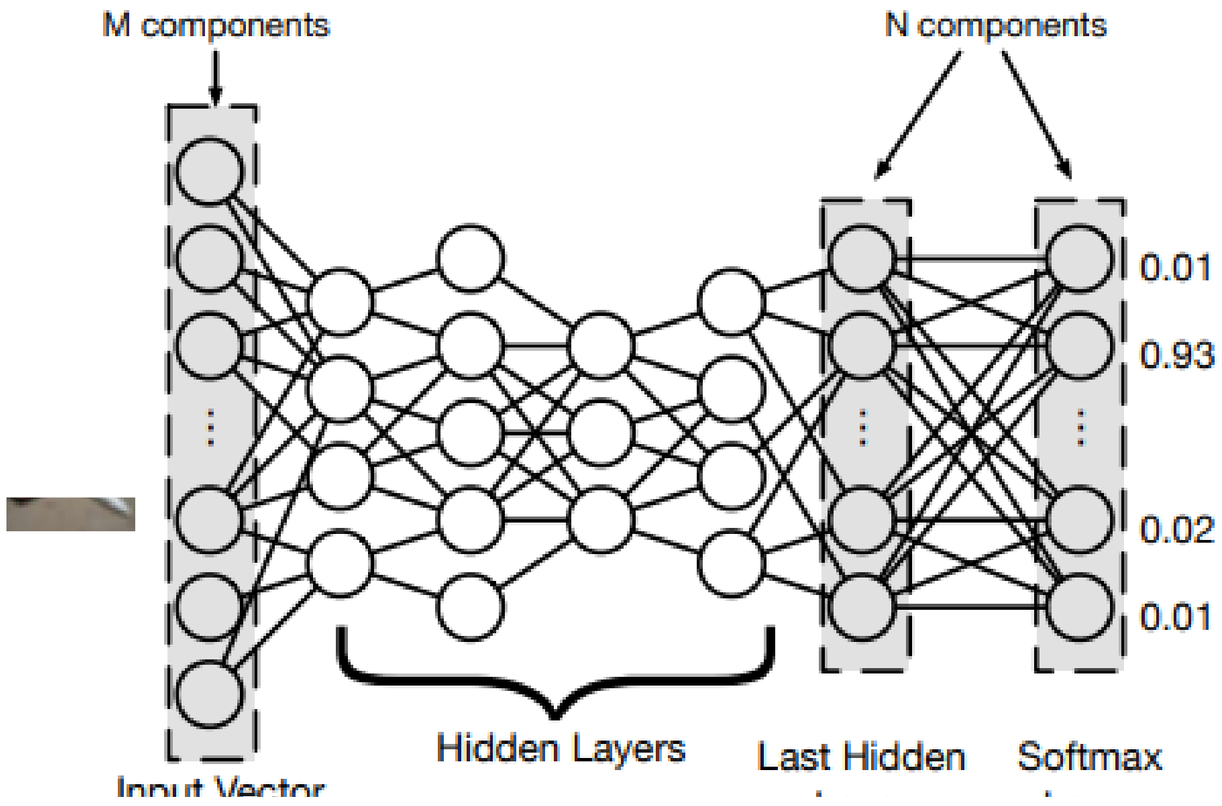
Implementation : Impacts Baseline model in Presence of FGSN

At Inference

Adversarial image

Input : Test set +
Add perturbations

ϵ is increased from 0 to 0.3



max the loss

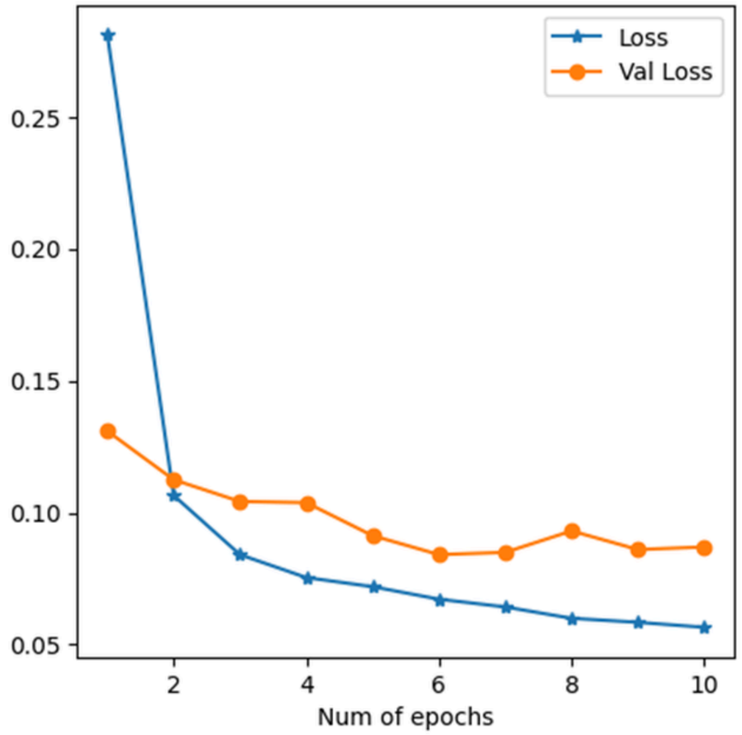
Test accuracy dropped till 30 %

8

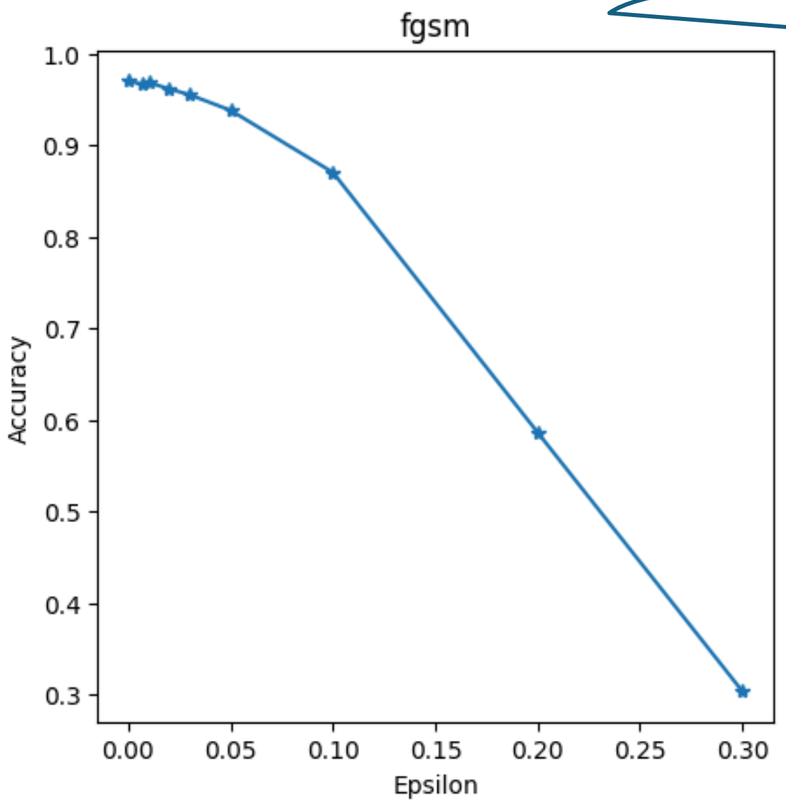
Misclassification

Results : In Presence of FGSM

Test accuracy dropped from 0.97 to 0.30 %



Variation of the train and valid loss with no of epochs for the training of the DNN



Effect of the FGSM attack on DNN for different noise levels in absence of defensive distillation

Inference on Test set :

Epsilon	No of correctly classified cases	Total no of cases	Classification accuracy
0	9708	1000	0.970
0.007	9668	1000	0.966
0.01	9688	1000	0.966
0.02	9619	1000	0.961
0.03	9554	1000	0.955
0.05	9384	1000	0.938
0.1	8705	1000	0.870
0.2	5863	1000	0.586
0.3	3034	1000	**0.303**

The effect of the FGSM attack on the classification performance of DNN

Implementation :CNN Architecture

Parameter	MNIST Architecture
Epochs	10
Learning rate	0.0001
Loss function	Cross entropy
Device	CPU , Google colab

Some training parameters

Distillation models:

Intuition : Transferring knowledge from larger architectures (Teacher) to smaller ones (Student).

- reduce size of DNN architectures (model compression) for resource constrained devices (e.g. smartphones)
- Robustness

Defense Distillation: DNN Architecture

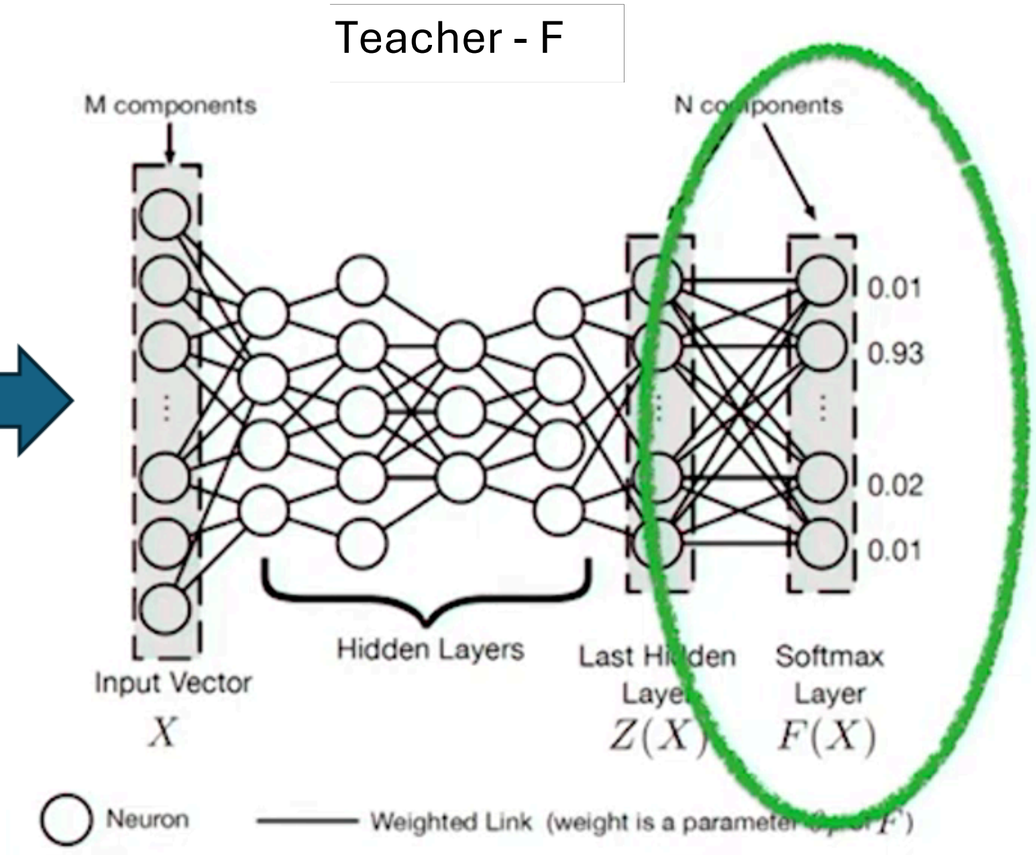
Main Idea : Instead of transferring knowledge between different architectures, use the knowledge extracted from a DNN to improve its **own** resilience to adversarial samples.

Teacher F – Student F1– CNN architecture

Train set image

Input

Teacher - F



T= 100

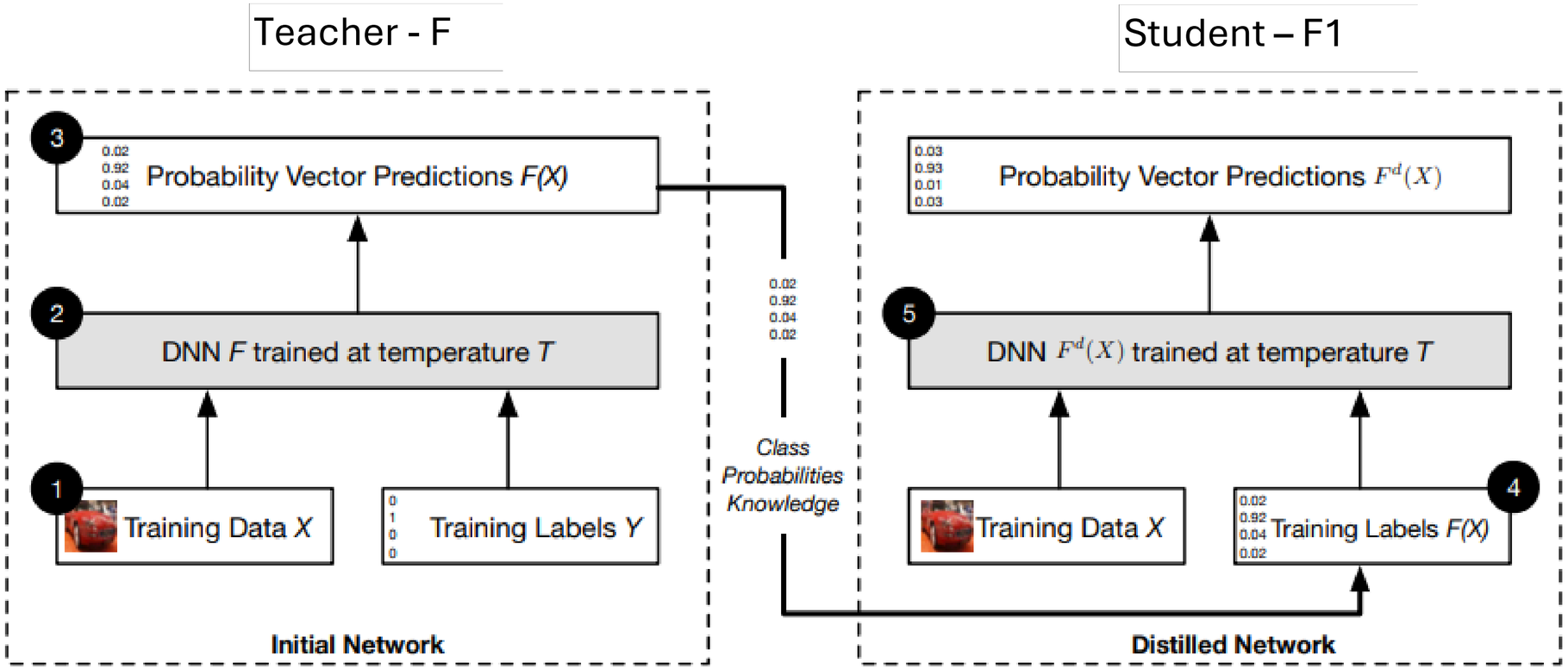
$$F(X) = \left[\frac{e^{z_i(X)/T}}{\sum_{l=0}^{N-1} e^{z_l(X)/T}} \right]_{i \in 0..N-1}$$

This output is used as input for training student network F1

Distillation as Defense: CNN architecture

Soft targets :

- Teacher's output probabilities [0.85, 0.05, 0.04, 0.03, 0.01] richer info than just the correct class.



Terms:

Softmax :

- raw logits → probability distribution [0.95, 0.02, 0.01, 0.01, 0.01] → Not informative beyond the top prediction.

Temperature (T) :

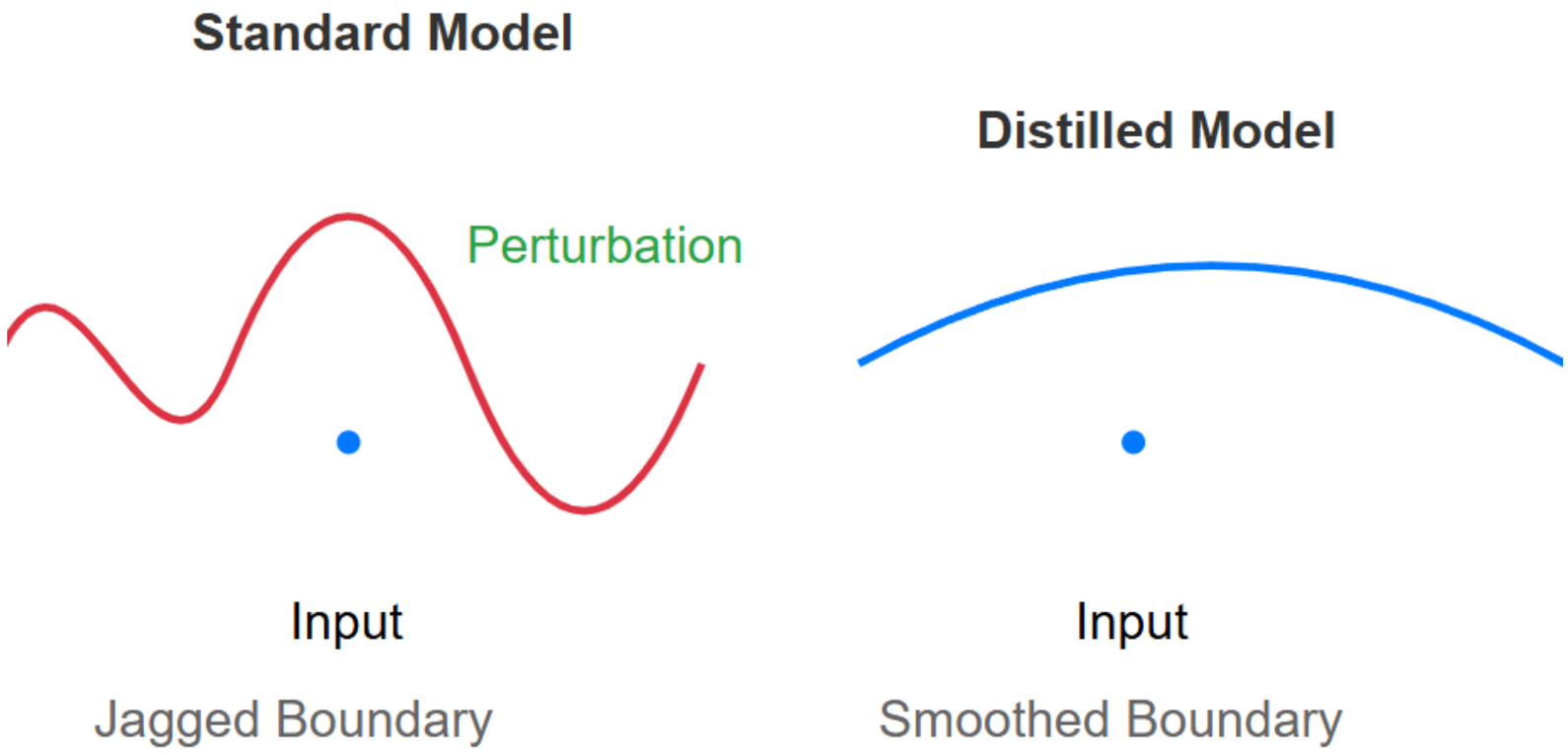
$$P_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

- Softens the probability c nax [0.85, 0.05, 0.04, 0.03, 0.01]

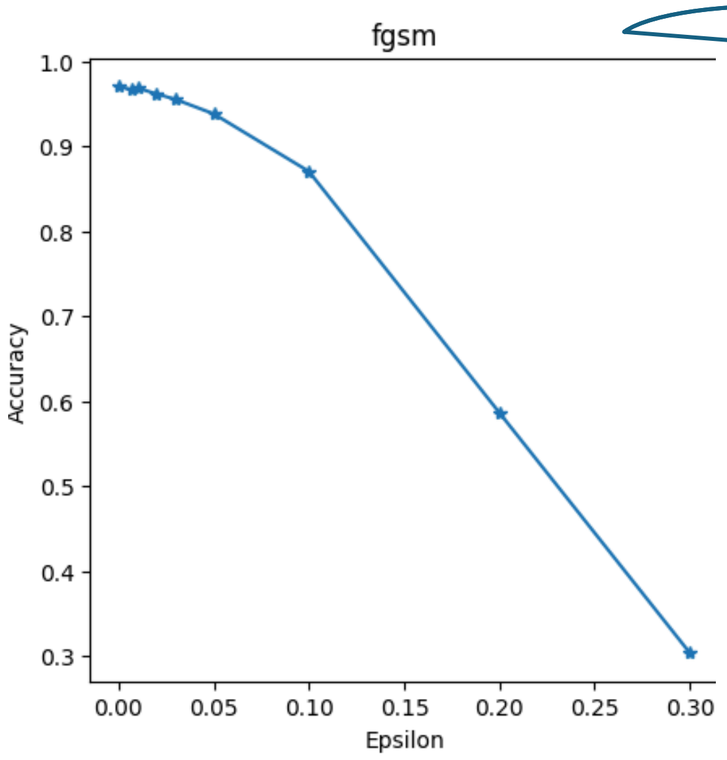
  - Higher T → softer, mo

$$P_i = \frac{exp(z_i/T)}{\sum_j exp(z_j/T)}$$

Why it works?


Standard Model
Perturbation
Input
Jagged Boundary
Distilled Model
Input
Smoothed Boundary

Results : In Presence of FGSM



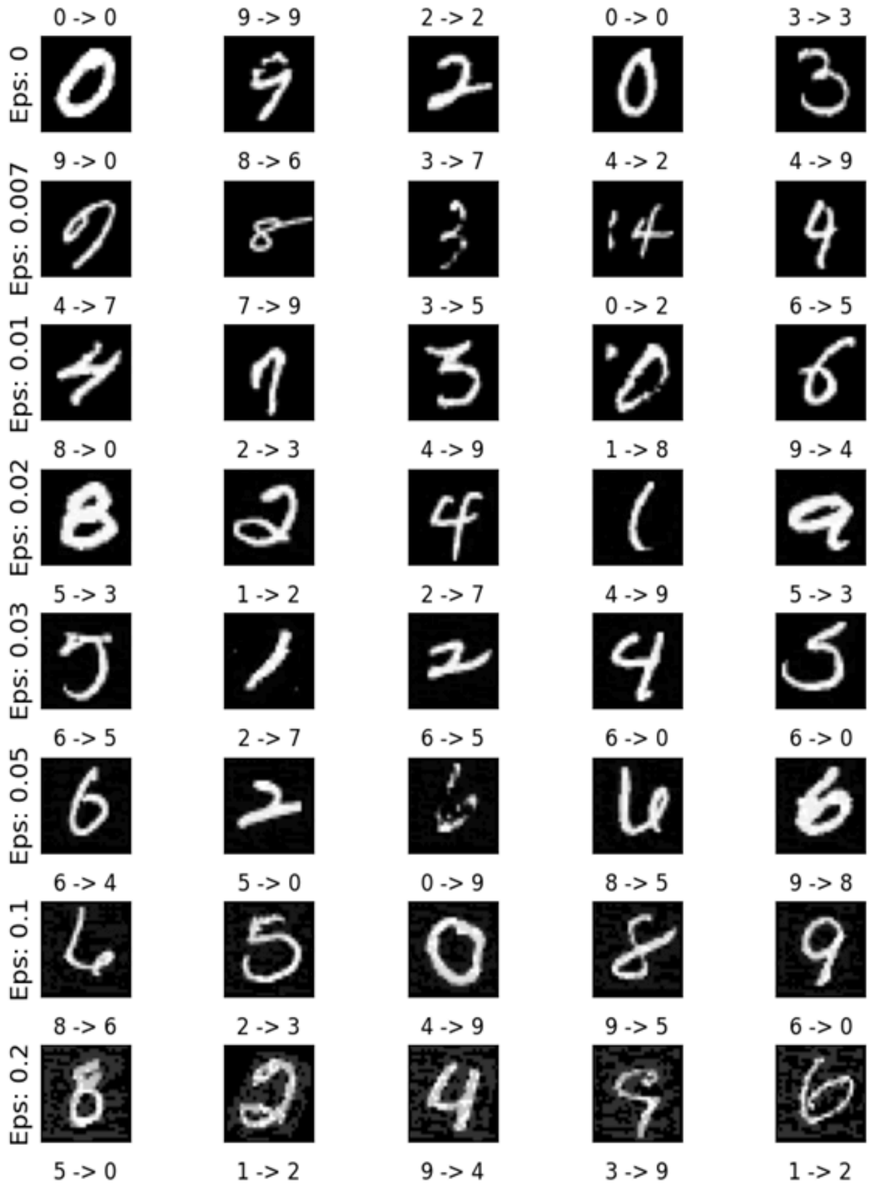
Effect of the FGSM attack on DNN for different noise levels in absence of defensive distillation

Inference on Test set :

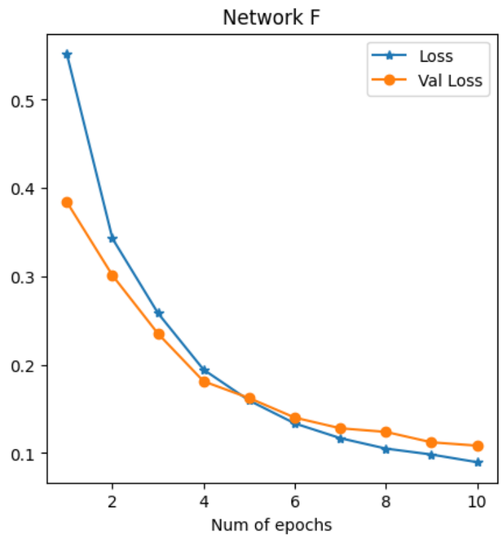
Epsilon	No of correctly classified cases	Total no of cases	Classification accuracy
0	9708	1000	0.970
0.007	9668	1000	0.966
0.01	9688	1000	0.966
0.02	9619	1000	0.961
0.03	9554	1000	0.955
0.05	9384	1000	0.938
0.1	8705	1000	0.870
0.2	5863	1000	0.586
0.3	3034	1000	**0.303**

The effect of the FGSM attack on the classification performance of DNN

Classification results under FGSM attack:

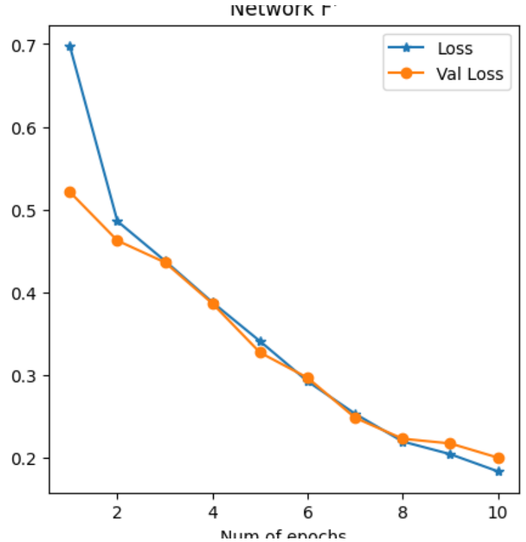


Results : FGSM in Presence of Distillation

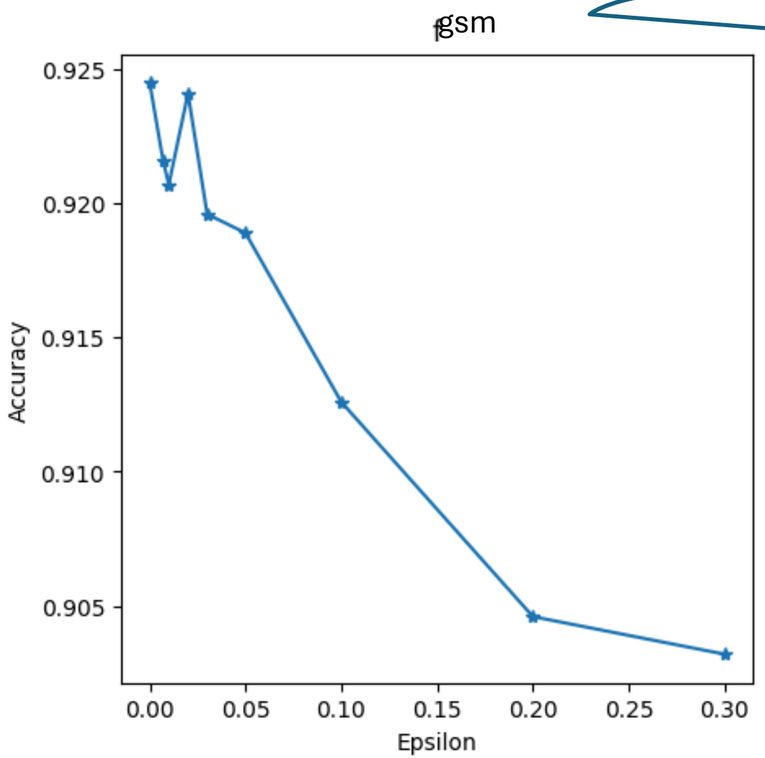


Teacher

Variation of the train and valid loss with no. epochs for training distilled DNN(T = 100)



Student



The effect of the FGSM attack on the classification performance of the distilled CNN model

Test accuracy only reduced by 2%

Inference on Test set :

Epsilon	No of correctly classified cases	Total no of cases	Classification accuracy
0	9245	1000	0.924
0.007	9216	1000	0.921
0.01	9207	1000	0.920
0.02	9241	1000	0.924
0.03	9196	1000	0.919
0.05	9189	1000	0.918
0.1	9126	1000	0.912
0.2	9046	1000	0.904
0.3	9032	1000	**0.903**

The effect of the FGSM attack on the classification performance of the distilled CNN model

Results : FGSM in Presence of Distillation

Test accuracy only reduced by 2%

Inference on Test set :

Epsilon	No of correctly classified cases	Total no of cases	Classification accuracy
0	9245	1000	**0.924**
0.007	9216	1000	0.921
0.01	9207	1000	0.920
0.02	9241	1000	0.924
0.03	9196	1000	0.919
0.05	9189	1000	0.918
0.1	9126	1000	0.912
0.2	9046	1000	0.904
0.3	9032	1000	**0.903**

The effect of the FGSM attack on the classification performance of DNN

Observation : The distilled network seems to be robust in defending against the adversarial samples created by the FGSM attack. The classification accuracy of the distilled network even at 30% noise level is found to be as high as 0.903.

Thank you !

https://arxiv.org/abs/1511.04508